

Project Final Report for CS 175, Winter 2026

Project Team Name: NLPandas 🐼

Project Title: Truth or Silence

Project Topic: Hallucination Detection for Reliable Question Answering

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Github Repository: [truth_or_silence](#)

1. Project Summary

We investigate the problem of hallucination detection for reliable question answering by building a system that can choose to abstain instead of answering when the model is not confident. We implement and evaluate a baseline that always answers, and a selective answering method that abstains using a confidence margin computed from the LLM's next-token distribution. We evaluate on the TruthfulQA multiple-choice validation split using accuracy, coverage, and abstention tradeoffs.

2. Data Sets

We use the TruthfulQA dataset in its multiple-choice configuration, accessed through the HuggingFace datasets library using `load_dataset("truthful_qa", "multiple_choice")`. We use the validation split, which contains 817 total questions. For controlled experiments and runtime feasibility, we evaluate subsets of size 25, 100, 350, and 500 questions. The dataset is publicly available through HuggingFace, automatically downloaded and cached locally, and requires no additional labeling since ground-truth answers are already provided.

Each example contains a question, a list of answer choices, and a label vector indicating the single correct answer. We convert the correct index into a letter (A, B, C, ...) for evaluation. Tokenization is handled by the pretrained model. We verify that each answer letter tokenizes into a single token to ensure clean and reliable confidence-margin computation.

	Property	Value
1	Dataset	TruthfulQA (multiple_choice)
2	Split Used	Validation
3	Total Questions	817
4	Subset Sizes Tested	25, 100, 350, 500
5	Task Type	Multiple Choice
6	Correct Answers per Question	1
7	Tokenizer Vocabulary Size	~32,000 tokens

TruthfulQA is well-suited for our project because it contains factual questions designed to expose model misconceptions and hallucinations. Since the task is multiple-choice with one correct answer per question, it allows precise accuracy measurement and makes it ideal for studying selective answering and abstention behavior.

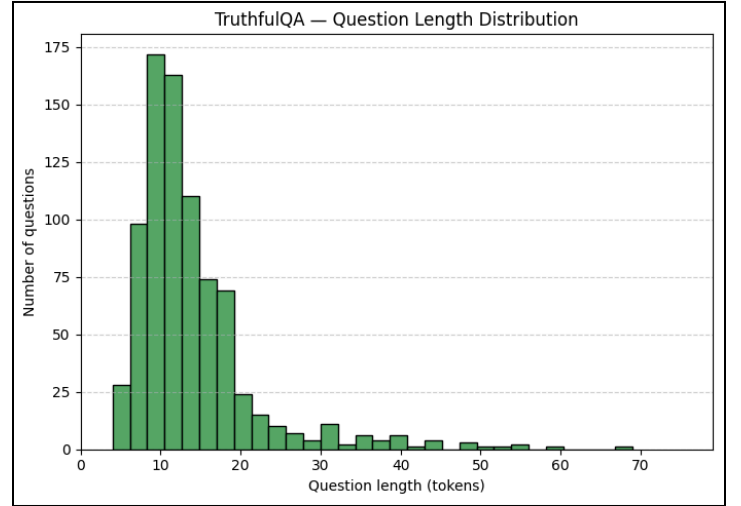
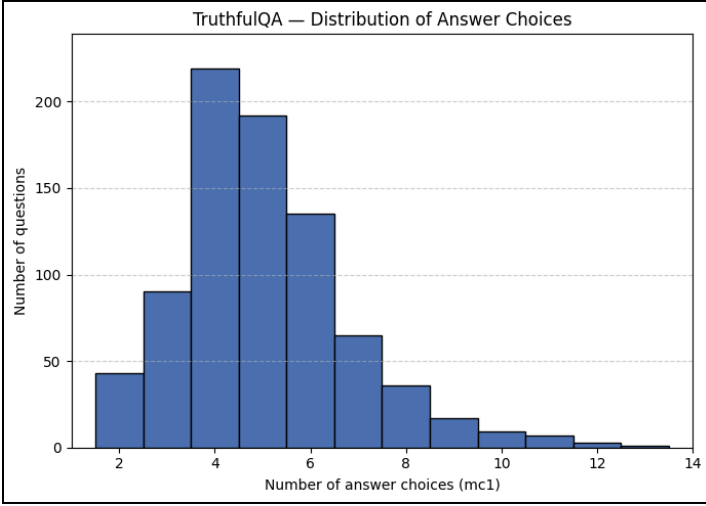


Figure 1: This histogram shows how many answer options each question has. It indicates the variability in multiple-choice difficulty and structure across the dataset.

Figure 2: This plot shows the distribution of question lengths in tokens. It helps characterize dataset complexity and the reasoning depth required by the model.

3. Technical Approach

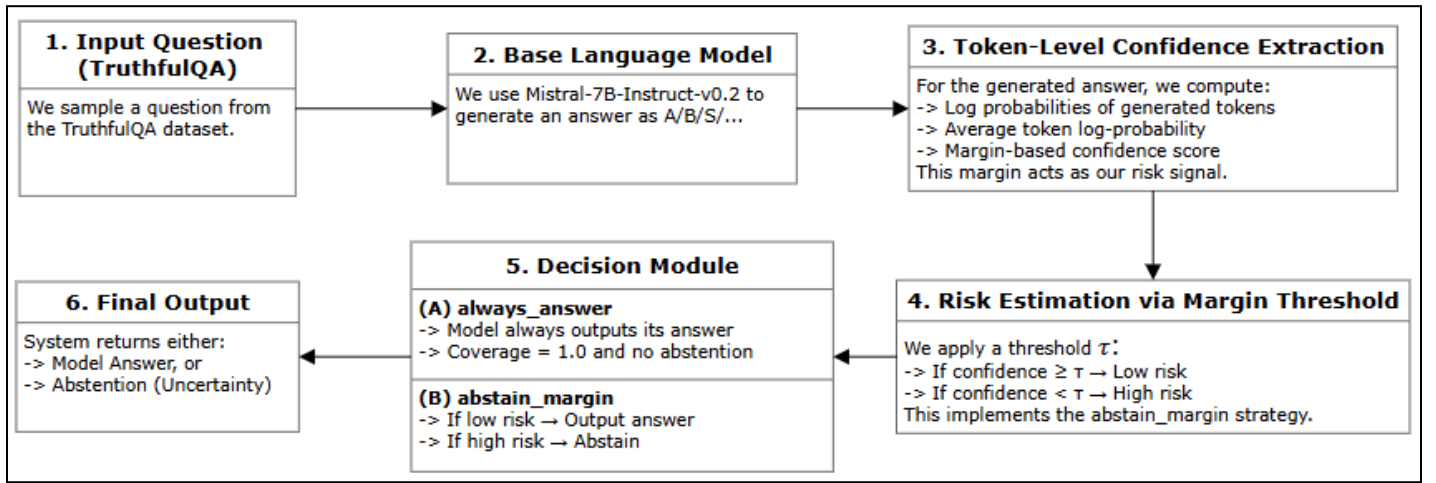


Figure 3: This figure shows our project’s Hallucination-Aware Question Answering Pipeline

Figure 3 shows our hallucination-aware question answering pipeline. Our project studies whether a language model can improve reliability by abstaining when it is not sufficiently confident in its answer. We use Mistral-7B-Instruct-v0.2 as the base model and evaluate it on the TruthfulQA multiple-choice benchmark. Because our task is multiple-choice question answering, we frame each prediction as a one-letter decision over the valid answer options. This lets us compare a standard always-answer baseline against confidence-based abstention methods in a clean and reproducible way.

3.1. Task framing: TruthfulQA Multiple-Choice as “One-Letter Prediction”

We use the TruthfulQA multiple-choice format. Each example includes:

- a question q
- a list of answer choices $[c1, c2, \dots]$
- a label vector that marks the correct option $[0, 1, \dots]$

Instead of generating a long answer, we design prompts that force the model to output only one letter corresponding to the selected option (A, B, C, ...). This makes evaluation clean and makes confidence scoring consistent, because we evaluate the next-token distribution over possible letters.

3.2. Prompt format used in experiments

We present the model with the question exactly as it appears in the dataset, followed by the multiple-choice options, and conclude the prompt with the instruction: “**Answer with ONLY the letter (A, B, C, ...):**”

This explicit instruction is important because it strongly constrains the model’s next-token prediction to be a single letter corresponding to one of the answer choices. By structuring the prompt this way, we ensure that the model’s next-token distribution directly reflects its preference among the available options. This makes our confidence margin calculation well-defined, stable, and aligned with the actual decision the model makes during generation.

3.3. Always-Answer Baseline (*always_answer*)

The baseline system always generates an answer for every question and never abstains.

Step-by-step Process:

1. Build the multiple-choice prompt explained above.
2. Prompt the model to generate exactly **one token** (the answer letter).
3. Extract the predicted letter using regex. We use regex as a precautionary measure to strip any additional characters like full stops or spaces (“A.”, “A_”, “(A)”) that may be generated by the model.
4. Compare the predicted letter to the correct answer.

This method has:

- **abstained = 0**
- **coverage = 1.0** (answers everything)

It provides a reference point: “What happens if the system is forced to answer every time?”

3.4. Confidence-Margin Abstention (*abstain_margin*)

We implemented an abstention mechanism called *abstain_margin*. Instead of always answering, the system first measures how confident the model is in its prediction and then decides whether to answer or abstain.

After generating an answer, we compute a confidence score using token-level log-probabilities from the model’s next-token distribution.

Margin Computation:

The threshold in our experiments is not on a 0–1 or 1–100 scale. It is applied to a log-probability margin, defined as:

$$\text{Margin} = \log P(\text{best letter}) - \log P(\text{second-best letter})$$

This margin measures how much more confident the model is in its top answer choice compared to the runner-up.

- A small margin means the model is unsure (top two options are close).
- A large margin means the model strongly prefers one option.

Threshold Scale Used:

We evaluated thresholds: $\tau \in \{0.5, 1.0, 2.0, 3.0, 5.0, 6.0, 8.0, 10.0\}$

Interpretation:

- Low threshold (0.5) → very small confidence gap required → model rarely abstains
- High threshold (10.0) → very large confidence gap required → model abstains more often

As the threshold increases:

- Abstention increases
- Coverage decreases
- Accuracy on answered questions typically increases

This wide range allows us to clearly observe the accuracy–coverage tradeoff.

Step-by-step Process:

For each question:

1. Build the multiple-choice prompt.
2. Run the model to generate exactly one token (the answer letter).
3. While generating, also ask the model to return the scores (logits) for that same step:
4. Convert those scores into log-probabilities.
5. From the full vocabulary, keep only the log-probabilities for the answer letters (A, B, C, ...).
6. Sort the letters by their log-probability.
7. Compute the margin, which is: $\text{Margin} = \log P(\text{best letter}) - \log P(\text{second-best letter})$

The margin tells us how much more confident the model is in its top choice compared to the next best option.

Decision rule:

- If $\text{margin} < \text{threshold } \tau \rightarrow \text{abstain}$
- If $\text{margin} \geq \text{threshold } \tau \rightarrow \text{answer}$

The system only answers when the top choice is clearly more likely than the second-best option.

3.5 Verbalized Confidence Abstention

We also implemented an alternative confidence method based on **verbalized confidence**. In this approach, the model is asked not only to choose an answer letter but also to state how confident it is on a 0–100 scale. The generated output is parsed into:

- a predicted answer letter
- a numeric confidence value

We then apply an abstention threshold directly to the self-reported confidence score. If the reported confidence is below the threshold, the system abstains; otherwise, it answers. We evaluated thresholds on the 0–100 scale, specifically:

$$\tau \in \{50, 60, 70, 80, 90, 95\}$$

This method was useful as a comparison because it tests whether the model’s own explicit confidence estimate is more informative than the token-level signal from its next-token distribution. Although this approach was fully implemented and evaluated, it did not perform as well as the token-margin method and was therefore not selected as the final approach.

3.6 Methods Considered but Not Pursued Further

We also considered a sampling-based confidence strategy in which the model would generate multiple responses per question and confidence would be inferred from consistency across samples. However, after discussion with the TA, we decided not to pursue a naïve one-letter Monte Carlo version because repeated greedy or near-greedy sampling would likely return the same highest-probability answer in most cases, providing limited additional information. Given the project timeline, we chose instead to deepen our analysis of the token-margin method and compare it against verbalized confidence.

3.7 Final Technical Approach Used in the Report

Although we explored multiple confidence strategies, the main method used in our final report is the token log-probability margin abstention model, compared against the always-answer baseline. The verbalized-confidence method is included as an explored alternative, but not as the final system, because its self-reported confidence scores did not align as well with correctness in practice. This comparison helped us better understand which types of confidence signals are useful for hallucination-aware question answering.

4. Experiments and Evaluation

We evaluate our final system on the 500-question subset of the TruthfulQA multiple-choice validation set, following the TA’s suggestion to focus on the largest and most stable sample. For each question, Mistral-7B-Instruct-v0.2 is prompted to output a single answer letter. We then extract the logits for that same generation step, apply log-softmax to obtain log-probabilities, restrict scoring to the valid answer letters for that question, and compute a confidence margin defined as the difference between the top-1 and top-2 answer-letter log-probabilities. A larger margin indicates that the model strongly prefers one option over the runner-up, while a smaller margin indicates uncertainty. The abstention rule is simple: if the margin is below threshold τ , the system abstains; otherwise, it answers. We compare this selective answering system against an always-answer baseline and report answered-question accuracy, coverage, abstained count, and error-overlap analyses.

The main result from our evaluation is a clear accuracy-coverage tradeoff. On the 500-question subset, the always-answer baseline reaches 0.8428 accuracy. Under margin-based abstention, answered-question accuracy rises from 0.8286 at $\tau = 0.5$ to 0.9031 at $\tau = 10.0$, while coverage falls from 0.992 to 0.846. As shown in *Figure 4*, increasing the threshold makes the system more conservative: it answers fewer questions, but the answers it keeps are more reliable. This is the central performance result of the project.

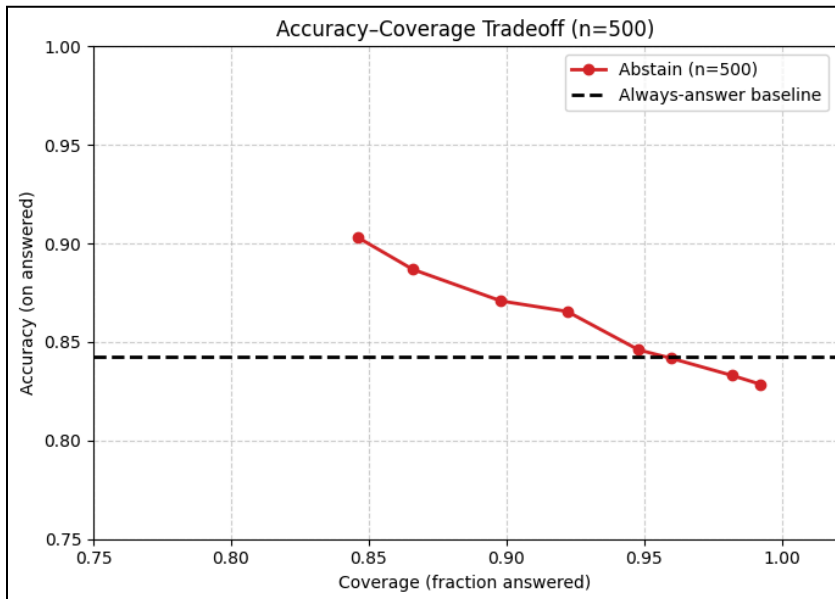
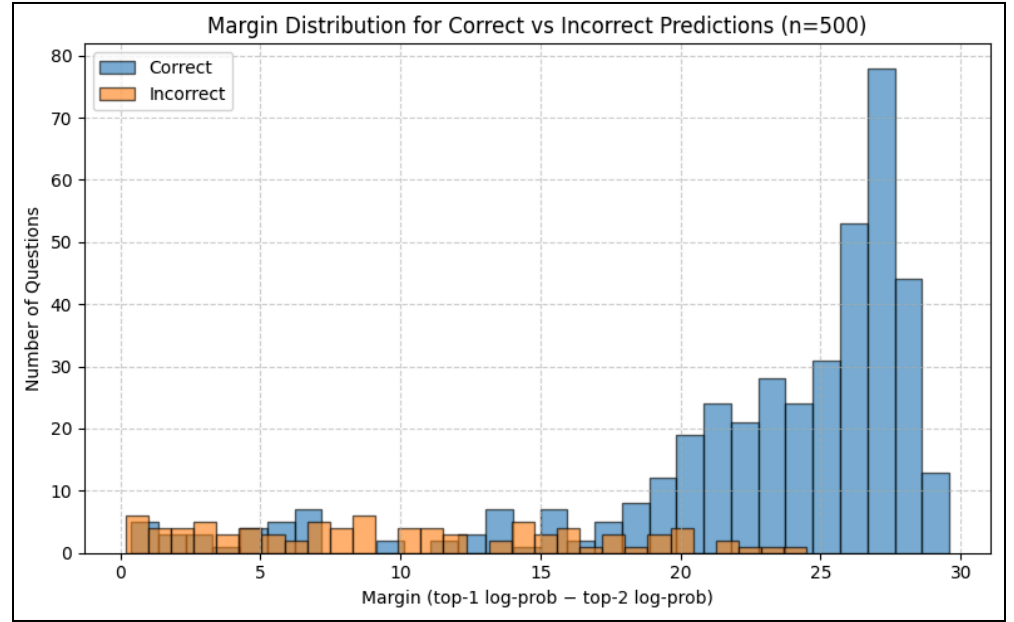


Figure 4: Accuracy-coverage tradeoff for the token-margin abstention model on the 500-question TruthfulQA subset. As the threshold increases, confidence decreases but answered-question accuracy rises, showing that abstention improves reliability at the cost of answering fewer questions.

To understand why abstention works, we analyzed the margin itself as a confidence signal. We observed that correct predictions have much larger margins than incorrect predictions. On the 500-question sample, correct answers have mean margin 22.86 and median 25.27, while incorrect answers have mean margin 9.99 and median 8.75. As shown in *Figure 5*, this separation indicates that the top-1 vs. top-2 log-probability gap is not just a heuristic, but a meaningful indicator of confidence. In effect, the abstention system is using a signal that already correlates with correctness, which explains why filtering low-margin cases improves reliability.

Figure 5: Distribution of token log-probability margins for correct/incorrect predictions. Correct answers have a larger top-1 vs. top-2 gaps than incorrect answers, showing that the margin is a meaningful confidence signal.



We also examined whether abstention is selectively filtering risky outputs or merely discarding answers at random. The results show that it is genuinely catching many of the baseline’s mistakes. Among the 98 questions answered incorrectly by the always-answer baseline, the abstention system catches only 3 at $\tau = 0.5$, but catches 57 at $\tau = 10.0$, corresponding to 58.2% of baseline errors. As seen in *Figure 6*, this is strong evidence that the detector is meaningfully targeting hallucination-prone cases. At the same time, the gains are not free because stricter thresholds also remove some correct answers, which is the cost of higher reliability.

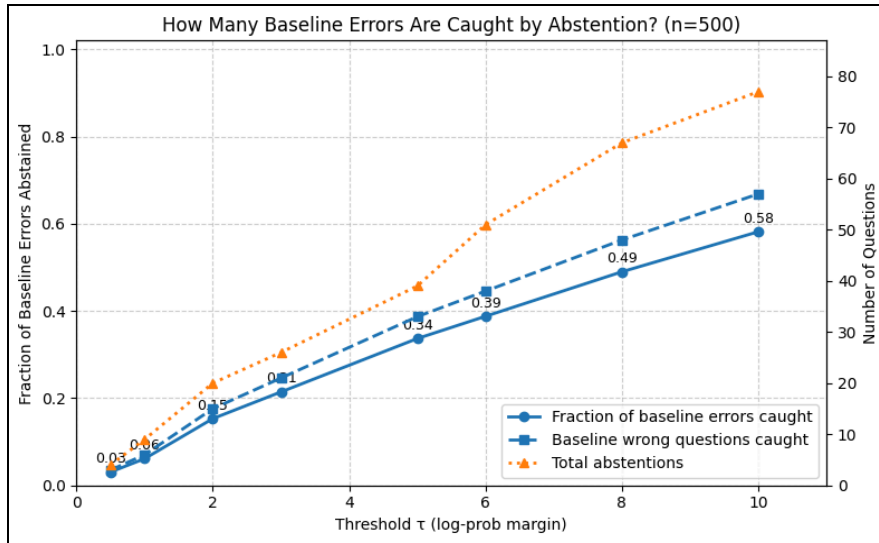
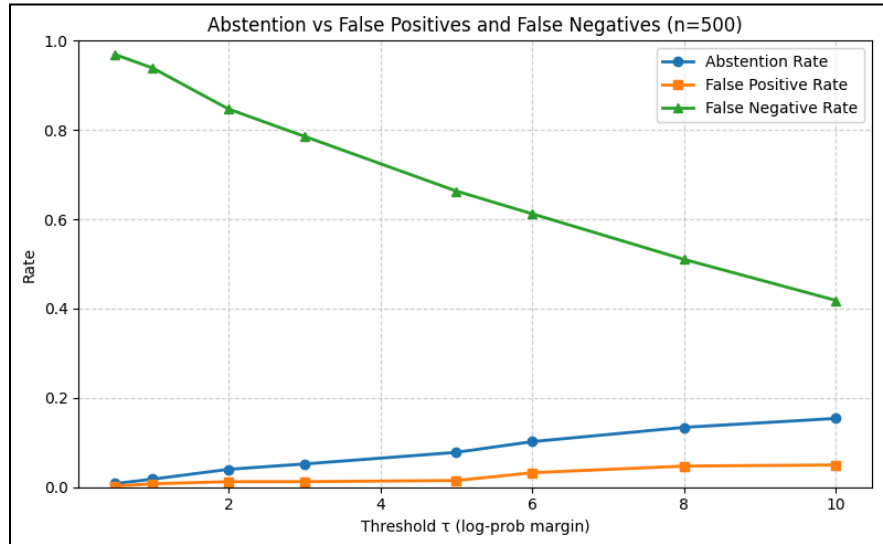


Figure 6: Fraction of always-answer baseline errors caught by the token-margin abstention system as the threshold increases. Higher thresholds filter out a larger share of risky baseline mistakes.

A closer analysis revealed two important failure modes that limit the reliability of the margin-based method. The first is the false positive case, where the detector abstains even though the baseline answer would have been correct. These are underconfident correct answers and reduce coverage unnecessarily. The second is the false negative case, where the detector still answers a question incorrectly instead of abstaining. These are overconfident errors and are more serious because they remain in the answered set as hallucinations. At threshold $\tau = 5.0$, the mean margin is 1.55 for false positives, 2.39 for true positives, 12.87 for false negatives, and 23.67 for true negatives. This pattern shows that the margin signal is informative but imperfectly calibrated: abstained cases usually do have low margins, but some incorrect

answers still receive surprisingly high confidence. *Figure 7* further shows that as the threshold increases, the false negative rate drops much faster than the false positive rate rises, meaning stricter abstention substantially reduces missed hallucinations while introducing relatively few unnecessary abstentions on correct answers.

Figure 7: False-positive and false-negative behavior of the token-margin detector across thresholds. As the threshold increases, the false negative rate drops much faster than the false positive rate rises, indicating that stricter abstention removes many missed hallucinations while introducing relatively few unnecessary abstentions.



We then investigated what might be causing these remaining false cases. The strongest structural trend appears to be question length. As question length increases, the average margin decreases substantially: from 22.90 for questions in the 0–10 token bin to 11.76 for questions in the 31+ token bin. At the same threshold, abstention rate also rises from about 0.05 to 0.25 across those bins. As shown in *Figure 8*, this suggests that longer questions tend to make the model less certain, likely because they require more reasoning or present more semantic complexity. We also analyzed the number of answer choices and found suggestive but noisy evidence that questions with more choices may increase abstention, but the bins with many choices are sparse, so we do not over-interpret that pattern. Overall, the clearest finding is that question difficulty, especially in the form of longer questions, compresses the margin and contributes to detector uncertainty.

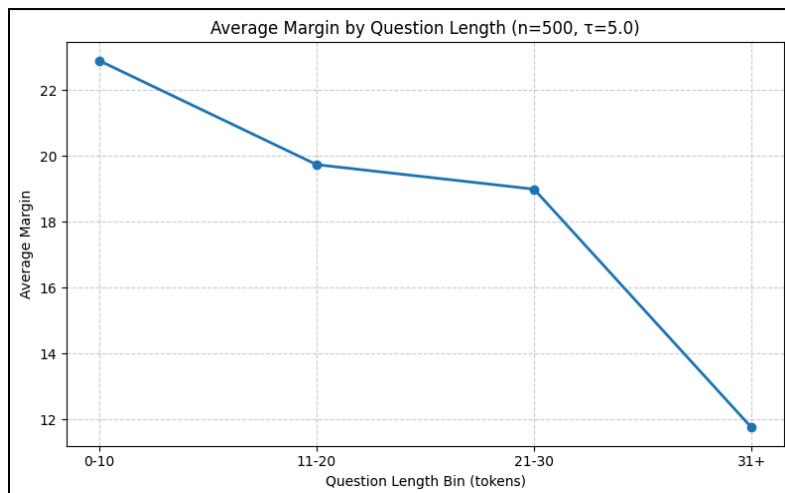
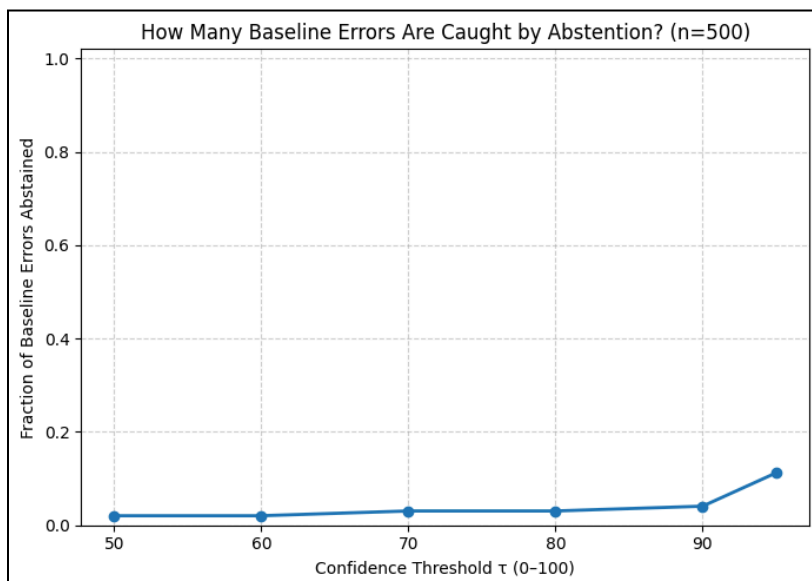


Figure 8: Average token margin confidence by question length. Longer questions tend to have lower average margins, suggesting that the increased question complexity makes the model less certain

During our experimentation, we also considered alternative methods to compute confidence to see if that would improve results. First, we tested an alternative confidence signal based on verbalized confidence, where the model outputs both an answer letter and a self-reported confidence score from 0 to 100. This method performed substantially worse than the token-margin approach. On the same 500-question set, answered-question accuracy only ranged from about 0.7796 to 0.7879 across thresholds 50 to 95, while the always-answer baseline remained 0.8428. As shown in *Figure 9*, the verbalized-confidence method catches only a small fraction of baseline errors even at its strictest threshold, which further

confirms that self-reported confidence does not align well with correctness in this setting. Because the self-reported confidence signal failed to produce the strong reliability gains seen with token log-probability margins, we did not pursue it further.

Figure 9: Fraction of baseline errors caught by the verbalized-confidence abstention method. Compared with the token-margin approach, self-reported confidence catches relatively few risky errors, indicating that verbalized confidence is a weaker signal for reliability in this setting.



We also planned to explore a sampling-based confidence method in which the model would generate multiple responses per question and confidence would be inferred from consistency across samples. However, after discussion with the TA, we concluded that naïvely repeating one-letter generation would likely produce limited new information, because the same highest-probability letter would often dominate all samples. Given the time constraints of the final project, we decided not to spend effort implementing a weak version of this method and instead focused on deeper analysis of the token-margin model, which yielded more interpretable findings about detector failures and reliability.

5. Lessons Learned and Insights

Our main takeaway is that the token-level log-probability margin is a useful and interpretable confidence signal for selective answering. Correct predictions tend to have much larger top-1 vs. top-2 gaps than incorrect predictions, and this separation is what makes abstention effective. The model does not become more knowledgeable under abstention; instead, it becomes better at withholding answers on cases where its internal preference structure is weak. That distinction was one of the most important conceptual lessons of the project.

A second insight is that abstention improves reliability in a meaningful way, not just cosmetically. As the threshold increases, the system filters an increasing fraction of baseline mistakes, eventually catching more than half of them on the 500-question sample. This confirmed our expectation that uncertainty estimates can help reduce hallucinations even without retraining the model. At the same time, the tradeoff is unavoidable, as higher reliability comes at the cost of lower coverage. Designing a useful system therefore means choosing a threshold that balances safety and usefulness rather than optimizing only one metric.

The most important limitation we observed is overconfident error. Some incorrect answers still receive high margins and survive the abstention filter. These false negatives are harder to fix than underconfident correct answers because they indicate that the model is internally confident in a wrong option, not merely uncertain. This suggests that confidence estimation alone may not be enough to eliminate hallucinations; calibration remains a central challenge. In other words, the margin helps, but it is not a complete solution.

We also learned that not all confidence signals are equally useful. The verbalized-confidence method was appealing because it asks the model directly how sure it is, but in practice that self-reported score was much weaker than the token-level signal. This was a valuable negative result: it showed that the model’s explicit confidence is less trustworthy than the structure of its next-token distribution. That finding helped us narrow our focus and invest more effort in deeper analysis of the margin-based system instead of chasing a weaker method.

Finally, our analyses suggest that the remaining reliability challenges are tied partly to question difficulty. Longer questions tend to have lower margins and higher abstention rates, and questions with more answer choices may show a similar effect, though less cleanly. For future work, one promising direction would be to explore richer confidence methods that go beyond one-token decoding. For example, a customized sampling-based method or a chain-of-thought-based confidence approach could ask the model not only for an answer letter but also for reasoning, then measure consistency across multiple sampled responses. That extra reasoning may introduce enough variation to reveal uncertainty that is hidden in single-step letter prediction. Another direction would be to test larger or differently calibrated models to see whether overconfident hallucinations can be reduced more effectively.